

Analysis on Database Testing Technology of Computer Software Development

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Abstract- The current development of computer technology is relatively fast. In the process of computer application and promotion, computer software development is an important part of it. Computer software development itself has a strong system and complexity. Therefore, carrying out effective database testing during the development of computer software is a prerequisite for ensuring stable and safe operation of computer software. As a result, we should pay attention to the effective application of database testing technology. In this way, we not only need to strengthen the research of database testing technology in the process of computer software development, but we also need to analyze the composition of database testing technology. Only in this way can the test efficiency and the accuracy of the test results be guaranteed.

Keywords: *Computer Software; Development and Design; Database Testing Technology; Application Points*

I. COMPUTER SOFTWARE DEVELOPMENT AND DATABASE

A. Computer Software Development

Computer software development mainly refers to the process of establishing a software system based on the actual needs of users. Software development itself includes processes such as requirement analysis, functional design, implementation and testing. In the process of software development, programming language is a key part of realizing software functions. Under normal circumstances, developers will use relevant software development tools to complete computer software development work. Therefore, we can subdivide the software into two parts: computer system software and application software. The programs in the running process of the computer can be called software and programs, and the related files are the components of the software. At present, the level of computer software development is constantly improving. There are more and more requirements for various functions of computer software. As a consequence, software development is an important foundation for promoting the long-term development of the computer industry [1].

B. Database Testing Technology

1) Database

A database refers to a collection of associated data programmed in a certain format stored in the computer system memory. The database is the location where the computer's electronic files are stored. Users can use the database to complete data query, increase and decrease, and delete operations.

2) Database Test

Database testing is to complete the testing process of computer software system database structure, data table and data calling relationship in accordance with database design specifications. Any computer software application process requires a database. The database is an important core of computer software, because it can ensure that the computer software fulfills its functional requirements. We are supposed to strengthen the database testing operations in the computer software development process, so as to ensure the safe and stable operation of computer software.

In the process of database testing, the testing of database components is an important work content. Moreover, according to the different stages of software development, there will be certain differences in the analysis content. In the initial stage, we need to conduct a comprehensive analysis of the substance of the entire database, that is, to analyze the data and information needs of users. In this process, relevant staff can use the database logic model to achieve analysis purposes. At the second stage, we need to analyze the development needs of the business. The test is based on entity testing, mainly including view testing, table structure testing, storage testing, etc. Although database testing may seem simple, it is actually more complicated. If the test process is not strictly controlled, it may affect the test results. Therefore, testers must have relatively rich work experience and professional technical level to ensure the smooth completion of each test task.

When performing entity testing on a database, it is mainly based on view testing, table structure testing and storage testing. Using physical testing, we can discover in time whether there are problems with various functions in the software development process and whether the software functions are stable and reliable in various environments. Using physical testing, we can improve and optimize the software, and improve the efficiency and safety of the software. In addition, using physical testing we can solve the problem of data redundancy and improve the optimization effect of the software. After completing the database test, the tester needs to take effective measures according to the problems and vulnerabilities presented in the test results, so as to make up for the defects in the database [2].

II. IMPORTANCE OF DATABASE TESTING IN COMPUTER SOFTWARE DEVELOPMENT

Database testing in the process of computer software development is an important basis for ensuring the stable

operation of computer software. It also provides important technical support for data information collection and application. In the actual testing process, the technical expertise of technical personnel will directly affect the application level of database testing technology. In actual testing work, technicians may focus on computer code compilation and system software development and maintenance, but not the application of database testing technology. In consequence, in the process of computer software development, the application of database testing technology may have some shortcomings and defects. Therefore, we need to strengthen the application of database testing technology in the development of computer software. Only in this way can we promote the development of the computer industry. In addition, in the application process of database testing technology, we can guarantee the stability of computer software system development. Database testing technology will have a direct impact on the security of the software system. In practice, technicians need to correctly understand the application and classification of database testing technology. At the same time, they must use scientific and effective maintenance measures to promote the application of database testing technology to ensure the scientificity and rationality of computer software system testing. There are clear and clear methods and structures in database testing, and they have requirements for the form of the database. For example, in the first paradigm, each column in the database table structure is required to be an independent unit, and the same column cannot have repeated values. At this stage, in the database system, in addition to the first normal form, the second normal form data and the corresponding relationship requirements of the data table components, the third normal form database must not have requirements such as function transfer dependencies. These commands still have some problems in the software system development process of the system. We need to continuously test and upgrade the database, so as to optimize the computer application technology [3].

III. DATABASE TESTING TECHNOLOGY IN COMPUTER SOFTWARE DEVELOPMENT

A. Conformance Test

When testing database consistency, we need to proceed from the following three aspects to ensure the smooth progress of the test operation. First, test the primary key of the table. In the process of table primary key testing, we need to analyze and summarize the requirements of system design. All table primary keys have corresponding information recording requirements. In order to ensure the accuracy of the test results, we need to complete the table primary key consistency test. If the test results are not unique records, you need to reset the primary key of the table. Second, test the primary and foreign key relationship between the table and the table. In the application process of database testing technology, testing the key fields of primary and foreign keys and their degree is an important content to ensure that they meet the consistency requirements. However, most computers do not pay much attention to the testing of the primary and foreign key relationships between

tables and tables in the software development process, which may lead to ambiguities in the standard test and the accuracy of the test results. Third, delete the consistency test. In the process of testing the cascading table, the relevant staff must ensure the consistency requirements for the deletion of information data. If the user deletes the main table data information, the corresponding report data will be deleted accordingly, which is the deletion requirement for data consistency. It can not only improve the processing efficiency of data information to a large extent, but also can use computer software to delete and update data information in time. Because the internal consistency mechanism of the database can improve the efficiency of deleting information data. But if there is a problem with the consistency mechanism, it will have an impact on the efficiency of data processing.

B. Capacity Test

The main function of the database in the process of computer software development is to store and manage various data information. The larger the database capacity, the more data resources it can store and manage, and the higher its management efficiency. In the software development process, we need to test the capacity of the database so that we can accurately determine whether the database meets the needs of the software. When testing the database capacity, we need to record the amount of data, mainly the total number of bytes occupied by each field. The data volume of the data table is mainly the product of the recorded data volume and the number of records. The total amount of data in each data table is the actual capacity of the database. In the software development process, the database is an important basis for the realization of the entire software function, and the size of the database will directly affect the running effect of the software function. In addition to the size of the data table that will affect the capacity of the database, there are many factors in the computer software system that will affect the capacity of the database. For example, the stored procedures and system tables of various view data information will occupy database capacity to a certain extent. The data in the table in the database is basic data. Generally, the capacity of the system log file is about twice the reserved file capacity. In the process of testing the database capacity, the tool we use is generally a data factory. This testing tool can quickly generate test data and has an intuitive user interface. Using this test tool, the staff can construct more complex data relationships in the shortest time. Software technicians can use this tool to generate the correct test database. Moreover, the test tool can adapt to many types of databases, so it has strong application value [4].

C. Pressure Test

The staff should pay attention to the correctness of the database stress test software in the process of computer software development. The process of testing database stress is a non-functional test content. Therefore, relevant staff need to reasonably evaluate the response speed of the software, and the factors that affect the response speed of the software are shown in Figure 1.

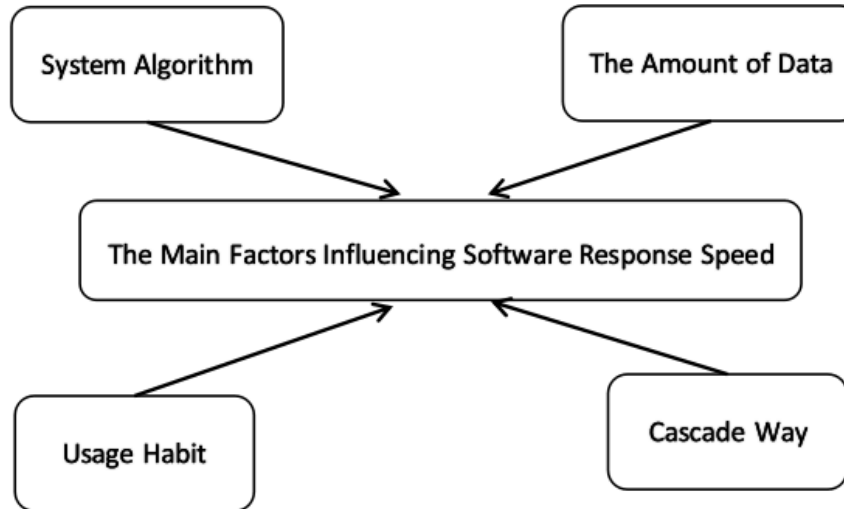


Figure 1. Factors Affecting Software Response Speed

The database stress test is mainly to test the response speed of the test software. If a large number of concurrent users access the software in a short period of time, the system functions of the software will be affected. In this case, the relevant staff must conduct a stress test. For example, a shopping website needs to test the flow pressure during the stress test. The main purpose of the stress test is to ensure the speed of the analysis software when the amount of data is relatively large. It can improve and upgrade the software to prevent web page paralysis. First of all, before applying the new software system, we must estimate the maximum amount of access that the software can withstand. Secondly, we can use the stress test to analyze the system pressure of the software in order to adjust and optimize the opening time and business interface of the webpage. Before the actual software system is applied, we need to carry out effective overhaul and maintenance of the software system, and carry out stress testing work to be able to accurately analyze the pressure that the software system can withstand. During stress testing, we can use professional testing tools to implement the testing process. When developing computers, we need to apply various database stress testing methods such as database testing and web page testing. Moreover, we can use a large number of tools to optimize the response time of the database to improve the response speed of the software system at runtime. When selecting an optimization tool, the relevant staff can use the Java project while using the system testing method in the integrated development environment to complete the testing process. In different software development environments and operating systems, technicians need to make a reasonable choice of optimization tools, so as to ensure the effectiveness of database stress testing.

D. Performance Testing

The database system may crash during the operation of the computer software system. This is mainly because the database system cannot withstand relatively large-scale information

calculations. Consequently, in the process of computer software development, technicians need to test the performance of the database. We must give full play to the actual functions of the database to ensure the stable operation of the software system. When developing computer software, we cannot use manual methods to complete system operation efficiency testing. Therefore, we need to use a professional software testing system to complete the performance testing process, so that we can accurately grasp the performance and functions of the database system. During the test, the capacity of each database is different. In order to ensure the optimization of database application performance, we need to effectively detect the software load and actual application scenarios. Only in this way can we rationalize and intelligently control software development. In the specific testing process, we need to ensure that the system operating environment is consistent. The development quality and application performance of computer software will have a certain impact on the overall performance of the computer. Especially when the system software has a stuck problem, if it cannot be solved in time, it will affect the stability of the computer system operation and is not conducive to improving customer experience. In order to ensure the operating performance of the software development system, technicians need to conduct self-checks on the performance of the system software regularly. On the basis of ensuring system performance, technicians can use advanced technical means to improve the processing capacity of data information. For example, in the process of applying FDatafactory technology, the data production capacity of the software system is relatively strong, and it can ensure the consistency and unity of data information. During the actual testing process, technicians need to carry out simulation training, improvement and upgrade of database technology, so as to give full play to the authenticity and reliability of computer software simulation. In the process of rapid development and improvement of digital technology, the future of the computer industry will develop faster, and the

application of information technology and the Internet of Things will be more closely linked, which will change people's lifestyles to a large extent. For this reason, we must attach importance to the stability of computer software development and application work, and must carry out reliable and effective database performance tests [5].

IV. CONCLUSION

All in all, in the current rapid development of computer technology, continuously improving the level of computer software development is an important measure to promote the development of the computer industry. Moreover, in computer development, we need to use database testing technology to find out the shortcomings and problems that exist before the software application in time, and optimize and improve the software, so as to improve the application level of the software.

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REFERENCES

- [1] Zhao Yifei. Research on Database Testing Technology of Computer Software Development[J]. Science and Technology Wind, 2019.
- [2] Niu Chuanming. Discussion on database testing technology of computer software development[J]. Journal of Zhejiang Institute of Water Resources and Hydropower, 2019.
- [3] Li Junmei. Discussion on database testing technology in computer software development[J]. China New Telecommunications, 2019, 21(09):164-165.
- [4] Wang Haiying. Research on database testing technology in software development [J]. Private Science and Technology, 2015, 000(004): 58-58.
- [5] Sun Huayou. Research on Database Testing Technology in Computer Software Development[J]. Journal of Changzhou Vocational College of Information Technology, 2017(6):25-27.